

FLIGHT OPERATIONS DOSSIER: DAL27 EMERGENCY TURNBACK

Ultra-Long-Haul Transpacific Sector | Augmented 4-Pilot Crew Protocol | Flight Telemetry & Rest Logs

FLIGHT IDENTIFIER Delta 27 (DAL27 / DL27)	AIRCRAFT TYPE Airbus A350-900 (A359)	ORIGIN AIRFIELD Atlanta (KATL)	PLANNED DESTINATION Seoul-Incheon (RKSI / ICN)
DEPARTURE TIME 11:35 PM EDT (03:35Z)	TURNAROUND POINT Hudson Bay / James Bay	DIVERSION DESTINATION Atlanta Return (KATL)	KATL GATE ARRIVAL 10:30 AM EDT (14:30Z)

SECTION 1: OPERATIONAL PROFILE & CREW COMPLEMENT

Delta Air Lines Flight 27 is an ultra-long-haul transpacific scheduled flight from Atlanta (KATL) to Seoul-Incheon (RKSI) with an operating block time of ~15 hours 20 minutes. Under FAA FAR Part 117 requirements, ultra-long-haul sectors exceeding 12 hours operating time require an **Augmented 4-Pilot Flight Deck Crew** consisting of:

- Pilot A (Captain):** Pilot-in-Command (PIC) - Designated Landing Pilot (Group 1)
- Pilot B (First Officer 1):** Senior FO - Designated Cruise Relief / Monitoring Pilot (Group 2)
- Pilot C (First Officer 2):** Relief FO - Designated Cruise Relief Pilot (Group 1)
- Pilot D (First Officer 3):** Relief FO - Designated Cruise Relief Pilot (Group 2)

4-PILOT DUTY STRUCTURE & REST ROTATION MECHANICS

To maintain continuous dual-pilot operation on the flight deck, pilots rest in dual pairs (2 active at controls, 2 off-duty in overhead crew rest berths). The cruise rest window is split into two equal operational halves, alternating between Group 1 (Pilots A & C) and Group 2 (Pilots B & D).

SECTION 2: CRUISE REST SCHEDULE & EMERGENCY EVENT MECHANICS

The scheduled cruise rest window spans 12 hours across the mid-flight sector. Below is the precise rest logging timeline leading directly up to the 03:30 AM EDT medical emergency over Hudson Bay:

TIME (EDT / LOCAL)	UTC TIME (Z)	REST ROTATION STAGE	FLIGHT DECK (CONTROLS)	REST FACILITY (BERTHS)
11:35 PM	03:35Z	Taxi & Departure: All pilots active	Pilots A, B, C, D	None
00:20 AM	04:20Z	Top of Climb (TOC): Cruise established at FL340	Pilots A & C (Group 1)	Pilots B & D (Group 2)
00:30 AM – 03:30 AM	04:30Z – 07:30Z	REST PERIOD 1 (3 hrs 00 min)	Pilots A & C (Active)	Pilots B & D (Off-Duty)
03:30 AM	07:30Z	EMERGENCY EVENT Crew Incapacitation	Pilot A & Pilot C	Pilots B & D Recalled
03:35 AM – 10:30 AM	07:35Z – 14:30Z	Emergency Diversion Transit: All pilots on deck	Pilots B & D (Flying)	Pilot C Monitoring / Med Care

EMERGENCY EVENT LOG (03:30 AM EDT / 07:30Z)

At 03:30 AM EDT, while operating over northern James Bay / Hudson Bay (~1,420 nm NNW of KATL), Pilot A (Captain) suffers sudden severe medical incapacitation while at the controls during Rest Period 1. Pilot C (active FO) initiates automated emergency holding, calls Pilot B and Pilot D from the rest facility, and issues Squawk 7700.

SECTION 3: FLIGHT TRACKER TELEMETRY & 180° TURNBACK PROFILE

The turnaround event produces a highly visible, dramatic telemetry imprint across global ADS-B tracking networks (FlightAware, Flightradar24):

EDT TIME	UTC TIME	FLIGHT MILESTONE / NAVIGATION EVENT	ALTITUDE	SPEED / SQUAWK
11:35 PM	03:35Z	Wheels up KATL Runway 27L. Climb via MOBIS SID.	0 \$ ightarrow\$ FL340	280 KIAS / 1542
00:20 AM	04:20Z	Level off cruise FL340 over Ohio. Rest Period 1 begins.	FL340	475 kts / 1542
03:30 AM	07:30Z	Incapacitation Event over James Bay: Transponder flips to Emergency.	FL360	480 kts / 7700
03:32 AM	07:32Z	180° U-Turn Execution: Sharp 25° bank turn southwest heading back to KATL.	FL360	440 kts / 7700
03:40 AM	07:40Z	Return cruise established heading 175° straight toward Georgia.	FL380	510 kts (Tailwind)
09:55 AM	13:55Z	Top of Descent (TOD) over northern Georgia / Contact Atlanta Center.	FL380 \$ ightarrow\$ FL100	320 KIAS / 7700
10:20 AM	14:20Z	TOUCHDOWN: KATL Runway 27R. Emergency vehicles roll.	0 ft AGL	142 KIAS / 7700
10:30 AM	14:30Z	GATE ARRIVAL: Gate F6. Medical team boards for extraction.	0 ft AGL	Taxi Speed